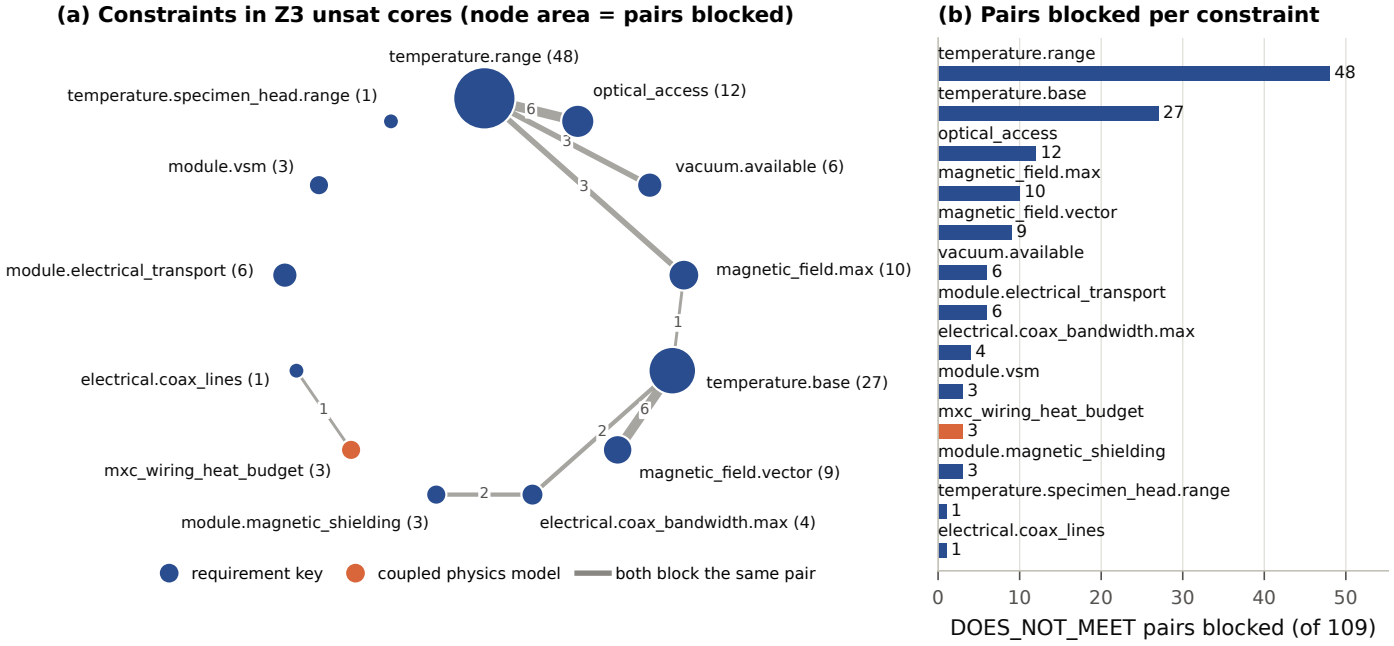
(a) Constraints in Z3 unsat cores (node area = pairs blocked)
temperature.range (48)
temperature.specimen_head.range (1)
optical_access (12)
module.vsm (3)
vacuum.available (6)
module.electrical_transport (6)
magnetic_field.max (10)
electrical.coax_lines (1)
temperature.base (27)
mxc_wiring_heat_budget (3)
magnetic_field.vector (9)
module.magnetic_shielding (3)
electrical.coax_bandwidth.max (4)
6
3
3
1
1
6
2
2
requirement key
coupled physics model
both block the same pair
(b) Pairs blocked per constraint
temperature.range
48
temperature.base
27
optical_access
12
magnetic_field.max
10
magnetic_field.vector
9
vacuum.available
6
module.electrical_transport
6
electrical.coax_bandwidth.max
4
module.vsm
3
mxc_wiring_heat_budget
3
module.magnetic_shielding
3
temperature.specimen_head.range
1
electrical.coax_lines
1
0
10
20
30
40
50
DOES_NOT_MEET pairs blocked (of 109)